Page 1 — Introduction to Computers

A computer is an electronic machine that accepts data, processes it according to instructions, stores information, and produces useful output. Modern computers can perform billions of operations quickly. Computers are used in education, business, science, communication, entertainment, healthcare, and engineering. The basic idea behind every computer system is the transformation of input into meaningful output.

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Page 2 — Computer Hardware

Computer hardware refers to the physical components of a computer system. Important components include the processor, memory, storage devices, motherboard, keyboard, mouse, display, and network hardware. The processor executes instructions, while memory temporarily holds information needed during execution. Storage devices keep data for longer periods. Together, these components allow software applications to operate.

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Page 3 — Operating Systems

An operating system is system software that manages computer hardware and provides services to application programs. Windows, Linux, and macOS are examples of operating systems. The operating system manages processes, memory, files, devices, security, and user interaction. It acts as an intermediary between applications and the underlying hardware.

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Page 4 — Data and Information

Data consists of raw facts, measurements, symbols, or observations. Information is data that has been processed and organized so that it becomes meaningful. Computers represent data using binary digits called bits. Eight bits form one byte. Larger units such as kilobytes, megabytes, gigabytes, and terabytes are commonly used to describe storage capacity.

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Page 5 — Computer Networks

A computer network connects devices so they can communicate and share resources. Networks may be small, such as a home network, or extremely large, such as the Internet. Common networking concepts include addresses, packets, routers, switches, protocols, and servers. Reliable communication requires agreed rules that determine how information is transmitted and interpreted.

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Page 6 — Databases

A database is an organized collection of data that can be stored, searched, updated, and managed efficiently. Relational databases organize information into tables containing rows and columns. SQL is commonly used to create queries and manipulate relational data. Databases are important in websites, banking systems, inventory management, universities, healthcare applications, and many other systems.

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Page 7 — Programming

Programming is the process of creating instructions that a computer can execute. Programming languages provide syntax and structures for expressing algorithms. Common languages include Python, Java, C++, JavaScript, and many others. A program can use variables, conditions, loops, functions, classes, and data structures to solve problems and automate tasks.

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Page 8 — Algorithms

An algorithm is a finite sequence of well-defined steps for solving a problem or completing a task. Good algorithms are usually evaluated using correctness, execution time, and memory usage. Searching, sorting, graph traversal, and optimization are common algorithmic problems. Understanding complexity helps developers choose solutions that remain efficient as input size increases.

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Page 9 — Artificial Intelligence

Artificial intelligence involves computer systems that perform tasks commonly associated with human intelligence, such as recognizing patterns, understanding language, making predictions, and supporting decisions. Machine learning is a major area of artificial intelligence in which systems learn patterns from data. Training data, model architecture, evaluation, and careful validation are important parts of machine learning.

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Page 10 — The Future of Computing

Computing continues to develop through advances in artificial intelligence, cloud computing, robotics, cybersecurity, quantum computing, and distributed systems. Future systems are likely to become more capable while also creating new challenges involving privacy, security, reliability, and responsible use. Learning fundamental concepts such as algorithms, operating systems, networks, databases, and programming provides a strong foundation for understanding these technologies.

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